

Data Preprocessing and Encoding

INTELLIGENT SYSTEMS

ESCUELA POLITÉCNICA DE INGENIERÍA DE GIJÓN



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Overview

Introduction

Data Preprocessing

Data Encoding

Summary

References

Introduction

Data Preprocessing

- Aims to improve data quality in order to increase the performance of machine learning models
- High-quality data usually leads to:
 - * Better generalization
 - * Faster learning
 - * More robust models
- Preprocessing typically includes:
 - * Better generalization
 - * Faster learning
 - * More robust models

Data Encoding

- Consists of adapting the data representation to the problem to be solved
- Two common scenarios:
 - * Converting categories into numbers
 - * Converting numerical values into categories
- Encoding is especially relevant in classification

Motivating Example

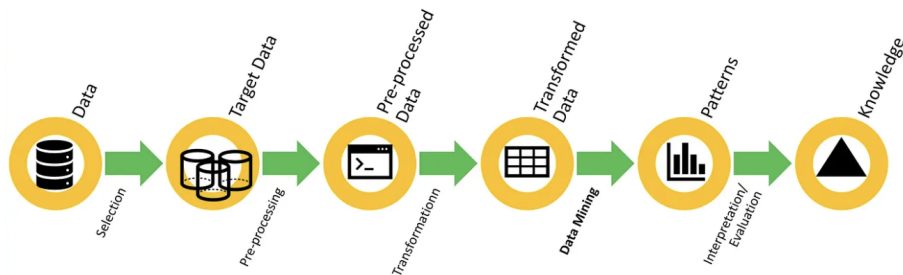
- Goal: determine whether a student passes a course based on academic data
- The evaluation rule is unknown, but historical data is available
- This is a typical supervised learning problem
- Available features: theory mark, lab mark, final outcome (pass / fail)

Student	ID	Passed?	Theory	Lab
A001	100001	False	8.0	0.04
A002	100002	False	10.0	0.35
⋮	⋮	⋮	⋮	⋮
A374	100374	False	7.6	0.40
A375	100375	True	4.0	0.90

Knowledge Discovery in Databases (KDD)

→ KDD defines the stages required to extract knowledge from data

→ Main phases:



Data Preprocessing

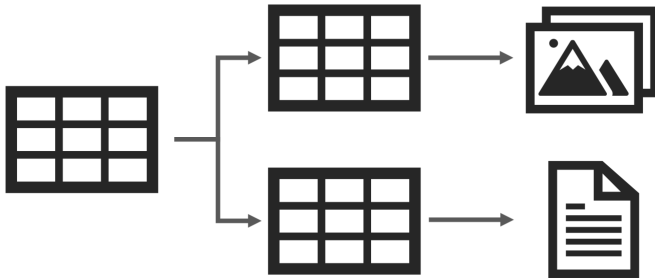
Data Preprocessing

Data Analysis

→ Data types:

- * Traditional data: numerical, textual
- * Multimedia data: images, video, audio

→ Most real-world datasets are **multimodal**, combining several data types



Data Analysis

→ Traditional data formats

Format	Description
.txt	Plain text files
.csv	Comma-Separated Values
.xlsx	Microsoft Excel spreadsheets
.json	JavaScript Object Notation
.pkl	Python Pickle format
.arff	Attribute-Relation File Format (Weka)
.sqlite	SQLite database

→ Multimedia data formats

Format	Description	Compression
.jpeg	Joint Photographic Experts Group	Yes
.png	Portable Network Graphics	Yes / No
.raw	Raw image data	No
.nifti	Neuroimaging Informatics Technology Initiative	Yes / No
.wav	Waveform Audio File Format	No
.mp3	MPEG-2 Audio Layer III	Yes

Data Preprocessing

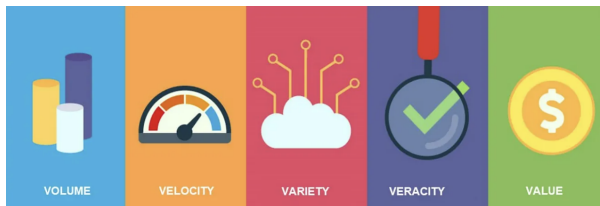
Data Volume and Sufficiency

→ Important questions:

- * Can the data fit in memory?
- * Do we need to split it into subsets?
- * Is a database system required?
- * Is the data streamed in real time?

→ Data sufficiency depends on problem complexity:

- * More complex problems require more data
- * Diversity is as important as quantity



Data Cleaning

→ Common issues:

- * Missing values
- * Incorrect or inconsistent values
- * Redundant or irrelevant variables

Student	ID	Passed?	Theory	Lab	Total
A001	100001	False	8.0	0.04	8.04
A002	100002	False	10.0	0.35	10.35
⋮	⋮	⋮	⋮	⋮	⋮
100374	False	7.6	0.40	8.00	NA
A375	100375	True	NA	0.90	4.9

Data Cleaning

→ Typical actions:

- * Imputation of missing values
- * Removal of erroneous samples
- * Feature selection

Passed?	Theory	Lab	Total
False	8.0	0.04	8.04
False	10.0	0.35	10.35
⋮	⋮	⋮	⋮
False	7.6	0.40	8.00
True	4.0	0.90	4.9

Data Cleaning

- Variables may be correlated or redundant
- Identifying these relationships improves learning
- Common techniques:
 - * Correlation analysis
 - * Information Gain
 - * Correlation-based Feature Selection

Passed?	Theory	Lab	Total
False	8.0	0.04	8.04
False	10.0	0.35	10.35
⋮	⋮	⋮	⋮
False	7.6	0.40	8.00
True	4.0	0.90	4.9

Data Cleaning

	Theory	Lab	Total
Theory	100%	-1%	73%
Lab	-1%	100%	68%
Total	73%	68%	100%



Passed?	Theory	Lab
False	8.0	0.04
False	10.0	0.35
⋮	⋮	⋮
False	7.6	0.40
True	4.0	0.90

Standardization and Normalization

- Many algorithms are sensitive to feature scales
- Feature scaling ensures that all variables contribute equally
- Two common techniques:
 - * Standardization
 - * Normalization

Standardization

→ Transforms data to have:

- * Mean equal to 0
- * Standard deviation equal to 1

$$z = \frac{x - \mu}{\sigma}$$

	Theory	Lab
Mean	0.32	-0.10
Std	0.78	0.84

⇒

Passed?	Theory	Lab
False	0.36	-0.97
False	0.87	-0.23
⋮	⋮	⋮
False	0.26	-0.11
True	-0.67	1.07

Normalization

→ Scales data to a fixed range, typically:

* $[0, 1]$

$$z = \frac{x - \min(x)}{\max(x) - \min(x)}$$

* $[-1, 1]$

$$z = 2 \cdot \left(\frac{x - \min(x)}{\max(x) - \min(x)} \right) - 1$$

Data Encoding

Data Encoding

- Adapting data to the problem we want to solve
- Typical scenarios
 - * Category $\rightarrow$ Number
 - * Number $\rightarrow$ Category
- Encoding is usually associated with classification problems
- It can be applied to:
 - * Input variables (x)
 - * Output variables (y)

Encoding: Category to Number

→ Key idea:

- * Machine learning models do not predict text or categories
- * They operate on numerical values.

→ Binary classification:

Passed?	Theory	Lab		Passed?	Theory	Lab
False	0.36	-0.97	⇒	0	0.36	-0.97
False	0.87	-0.23		0	0.87	-0.23
⋮	⋮	⋮		⋮	⋮	⋮
False	0.26	-0.11		0	0.26	-0.11
True	-0.67	1.07		1	-0.67	1.07

Yes → 1, No → 0

Encoding: Category to Number

→ Multi-class: One-Hot Encoding

* No intrinsic order between classes

Class	Theory	Lab		PL-1	PL-2	PL-4	Theory	Lab
PL-1	0.36	-0.97		1	0	0	0.36	-0.97
PL-2	0.87	-0.23		0	1	0	0.87	-0.23
⋮	⋮	⋮	⇒	⋮	⋮	⋮	⋮	⋮
PL-4	0.26	-0.11		0	0	1	0.26	-0.11
PL-2	-0.67	1.07		0	1	0	-0.67	1.07

Encoding: Category to Number

→ Multi-class: Ordinal Encoding

- * Continuous values can only be assigned if an order exists between the classes

→ Multi-label classification: One-Hot Encoding

Class	Theory	Lab		PL-1	PL-2	PL-4	Theory	Lab
PL-1, PL-2	0.36	-0.97		1	1	0	0.36	-0.97
PL-2	0.87	-0.23		0	1	0	0.87	-0.23
⋮	⋮	⋮	⇒	⋮	⋮	⋮	⋮	⋮
PL-4	0.26	-0.11		0	0	1	0.26	-0.11
PL-2, PL-4	-0.67	1.07		0	1	1	-0.67	1.07

Encoding: Number to Category

- Numerical variables can be discretized
- Useful when qualitative interpretation is needed
- From regression to classification

Theory	Lab	Group		Theory	Lab	Group
0.36	-0.97	7.8	⇒	0.36	-0.97	Notable
0.87	-0.23	5.0		0.87	-0.23	Pass
⋮	⋮	⋮		⋮	⋮	⋮
0.26	-0.11	0.8		0.26	-0.11	Fail
-0.67	1.07	2.6		-0.67	1.07	Fail

Data Encoding

Encoding: Number to Category

Theory	Lab	Group
0.36	-0.97	Notable
0.87	-0.23	Pass
$\vdots$	$\vdots$	$\vdots$
0.26	-0.11	Fail
-0.67	1.07	Fail



Theory	Lab	Excellent	Notable	Pass	Fail
0.36	-0.97	0	1	0	0
0.87	-0.23	0	0	1	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
0.26	-0.11	0	0	0	1
-0.67	1.07	0	0	0	1

Summary

Summary: Preprocessing

- Goal: improve data quality and, consequently, model performance
- Main processes:
 - * Analysis and cleaning
 - * Standardization and normalization

Summary: Encoding

- Goal: Adapt data representations to the problem to be solved
- Mainly applied to classification problems
- Common transformations:
 - * Category $\rightarrow$ Number
 - * Number $\rightarrow$ Category

References

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